

// Day 42 of 60

DISTRIBUTED SYSTEMS

CAP Theorem Explained

→ Swipe through all 6 slides



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The Impossible Triangle

Pick any two — never all three

In a distributed data store:

C = Consistency

A = Availability

P = Partition Tolerance

// You can guarantee

// only TWO at any time.



This is a theorem, not a guideline



Partition Tolerance (P)

The one you can't avoid

Networks WILL fail.

Servers WILL disconnect.

Partition Tolerance =
System keeps operating
during network drops

// In real systems,
// you MUST choose P.



Network partitions are not optional



So It's Really C vs A

The real choice you make

Since P is mandatory...

During a network partition:

Choose C → refuse stale answers

Choose A → answer with stale data

// You can't have both.

// Pick your trade-off.



CAP is a runtime trade-off, not a static one



CP Systems

Consistent + Partition Tolerant

Behavior on partition:

- Refuse to answer
- Wait for consistency

Examples:

- MongoDB
- HBase
- Zookeeper

// Use for: banking, payments,
// inventory, anything money



Better to fail than be wrong



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AP Systems

Available + Partition Tolerant

Behavior on partition:

- Always answer
- May serve stale data
- Reconcile later

Examples:

- Cassandra
- DynamoDB
- CouchDB

```
// Use for: feeds, DNS,  
// shopping carts, social
```



Better to be slightly wrong than offline



The Decision Matrix

How to choose for your app

Question: Which is worse?

(a) User sees old data
→ Pick CP

(b) User sees no data
→ Pick AP

// Banking → CP

// Twitter feed → AP

// E-commerce → AP

// Inventory → CP



Match the trade-off to the business cost



done = true

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